ESC103: Engineering Mathematics and Computation

Exam Review

Brendan Tang

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1 Vectors

1.1 Vector Operations

Here are some basic vector operations in $\mathbb{R}^2$.

- Vector Addition: $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}$
- Scalar Multiplication: $c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$

Both are commutative, associative, and scalar multiplication is distributive over vector addition.

There are two ways to multiply vectors in $\mathbb{R}^3$. The dot product takes two vectors and returns a scalar. Here are two definitions for the dot product.

Definition. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$. The algebraic definition of the dot product is

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3.$$

Let θ be the angle between $\mathbf{v}$ and $\mathbf{w}$. The geometric definition of the dot product is

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \|\mathbf{w}\| \cos\theta.$$

The dot product is commutative, distributive over vector addition, and not associative.

The cross product takes two vectors and returns a vector that is orthogonal to both. Here are two definitions for the cross product.

Definition. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$. The algebraic definition of the cross product is

$$\mathbf{v} \times \mathbf{w} = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix}.$$

Let θ be the angle between $\mathbf{v}$ and $\mathbf{w}$. The geometric definition of the cross product is

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \|\mathbf{w}\| \sin\theta.$$

The cross product is anti-commutative, not distributive, and not associative.

1.1.1 Vector Projection

The normalization of a vector, $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

The scalar projection of $\mathbf{v}$ onto $\mathbf{w}$ is

$$\text{comp}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{w}\|} \text{ (algebraic)} \quad \text{comp}_{\mathbf{w}}\mathbf{v} = \|\mathbf{v}\| \cos\theta \text{ (geometric)}.$$

Combining both, the formula for the vector projection of $\mathbf{v}$ onto $\mathbf{w}$ is

$$\text{proj}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{w}^2\|}\mathbf{w}.$$

1.2 Vector Properties

Here are some special types of vectors in $\mathbb{R}^3$.

- Zero Vector: A vector with length 0, denoted by $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
- Unit Vector: A vector with length 1, denoted by $\hat{\mathbf{v}}$
- Standard Basis Vectors: Special unit vectors denoted by $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

The length of a vector $\mathbf{v}$ in $\mathbb{R}^3$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

Two vectors $\mathbf{v}$ and $\mathbf{w}$ are orthogonal if

$$\mathbf{v} \cdot \mathbf{w} = 0.$$

Two vectors $\mathbf{v}$ and $\mathbf{w}$ are collinear if

$$\mathbf{v} \times \mathbf{w} = \mathbf{0}.$$

1.3 Plane Geometry

A line in $\mathbb{R}^3$ can be represented by the vector equation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + a\mathbf{d} \quad a \in \mathbb{R},$$

where $\mathbf{d}$ is a direction vector and (x_0, y_0, z_0) is a point on the line.

Similarly, a plane in $\mathbb{R}^3$ can be represented by the vector equation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + a\mathbf{d}_1 + b\mathbf{d}_2 \quad a, \in \mathbb{R},$$

where $\mathbf{d}_1$, $\mathbf{d}_2$ are non-collinear direction vectors and (x_0, y_0, z_0) is a point on the plane.

A plane in $\mathbb{R}^3$ can also be represented by the scalar equation

$$ax + by + cz = d,$$

where $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ is a vector normal to the plane. (why?).

1.3.1 Interlude (Row and Column Pictures)

The row picture of a system of linear equations treats each equation as a geometric line or plane, and the solution is the point where they all intersect.

The column picture of a system of linear equations views the system as a linear combination of column vectors, and to solve you need to find the correct scalar multiple of each vector.

2 Matrices

2.1 Matrix Operations

Here are some basic matrix operations.

- Matrix Addition: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \end{bmatrix}$
- Scalar Multiplication: $c \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} ca_{11} & ca_{12} \\ ca_{21} & ca_{22} \end{bmatrix}$

Both are commutative, associative, and scalar multiplication is distributive over matrix addition.

If $\mathbf{A}$ is an $m \times r$ matrix and $\mathbf{B}$ is an $r \times n$ matrix, we can perform matrix multiplication $\mathbf{AB}$ to get an $m \times n$ matrix.

Definition. Let $\mathbf{A} = \begin{bmatrix} — \ \mathbf{a}_1 \ — \\ \vdots \\ — \ \mathbf{a}_n \ — \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_2 \\ | & & | \end{bmatrix}$. Then the row-column definition is

$$\mathbf{AB} = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{b}_1 & \cdots & \mathbf{a}_1 \cdot \mathbf{b}_n \\ \vdots & & \vdots \\ \mathbf{a}_n \cdot \mathbf{b}_1 & \cdots & \mathbf{a}_n \cdot \mathbf{b}_n \end{bmatrix},$$

where the value in the ith row and jth column of the resultant matrix is equal to $\mathbf{a}_i \cdot \mathbf{b}_j$.

Let $\mathbf{A} = \begin{bmatrix} | & & | \\ \mathbf{a}_1 & \cdots & \mathbf{a}_n \\ | & & | \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} — \ \mathbf{b}_1 \ — \\ \vdots \\ — \ \mathbf{b}_n \ — \end{bmatrix}$. Then the column-scaling definition is

$$\mathbf{AB} = \begin{bmatrix} b_{11} \begin{bmatrix} | \\ \mathbf{a}_1 \\ | \end{bmatrix} + \cdots + b_{n1} \begin{bmatrix} | \\ \mathbf{a}_1 \\ | \end{bmatrix} & \cdots & b_{1n} \begin{bmatrix} | \\ \mathbf{a}_n \\ | \end{bmatrix} + \cdots + b_{nn} \begin{bmatrix} | \\ \mathbf{a}_n \\ | \end{bmatrix} \end{bmatrix},$$

where the jth column of the resultant matrix is a linear combination of $\mathbf{a}_n$ using b_{nj} as coefficients.

Matrix multiplication is associative, distributive over vector addition, and not commutative.

2.1.1 Geometric Solutions to $\mathbf{Ax} = \mathbf{b}$

We can use row and column pictures to multiply a matrix by a vector. Consider the equation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

- Row Picture: We get three planes in $\mathbb{R}^3$,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + x_{32}x_2 + a_{33}x_3 &= b_3. \end{aligned}$$

- Column Picture: We get a linear combination of three vectors in $\mathbb{R}^3$,

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} + x_3 \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Question. *For each interpretation, what does it mean geometrically for there to be exactly one solution (x_1, x_2, x_3)? Infinitely many solutions? No solutions?*

2.1.2 Interlude (Important Definitions)

Here are some important definitions in linear algebra.

- Linear Independence: $\{\mathbf{a}_n\}$ are independent if the only solution to $\mathbf{Ax} = \mathbf{0}$ is $\{x_n\} = 0$
- Basis: A set of linearly independent vectors that span the vector space
- Span: The span of a basis is the set of all linear combinations of the basis
- Dimension: The dimension of a space is the number of vectors in any basis for that space
- Column Space: The column space of a matrix is the span of its column vectors
- Rank: The rank of a matrix is the dimension of its column space

2.2 Column-Row Factorization

Column-row factorization decomposes any $m \times n$ matrix $\mathbf{A}$ of rank r into the product of an $m \times r$ matrix $\mathbf{C}$ and an $r \times n$ matrix $\mathbf{R}$.